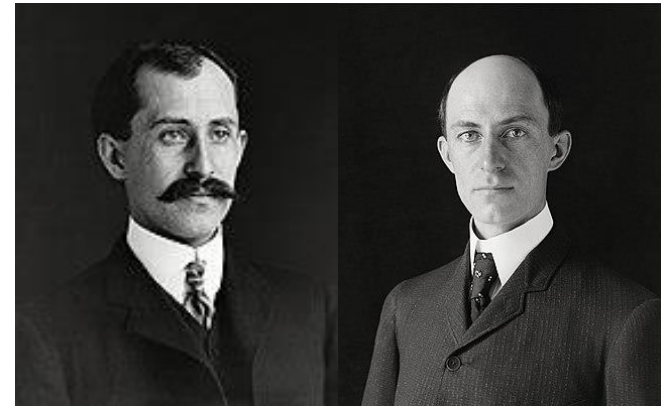
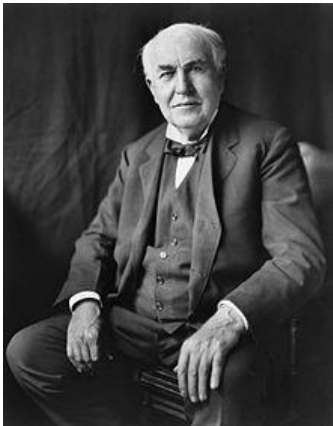


Design Thinking

Only Special People Can Innovate?



Any one can innovate?

Design Thinking

- A recipe (a step by step process) for innovation.
- Provides the tools you need to become an innovative thinker and uncover creative opportunities that are there – you're just not seeing them yet.
- It is the way designers think.
- An innovative problem solving approach.
- Human Centered Design

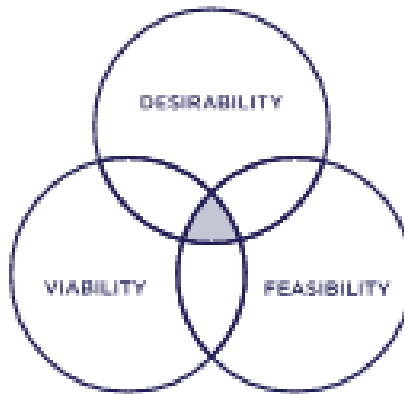
What is Design Thinking?

“Design thinking is a human-centered approach to innovation—anchored in understanding customer’s needs, [rapid prototyping](#), and generating creative ideas—that will transform the way you develop products, services, processes, and organizations.

By using design thinking, you make decisions based on what customers really want instead of relying only on historical data or making risky bets based on instinct instead of evidence”

-IDEO

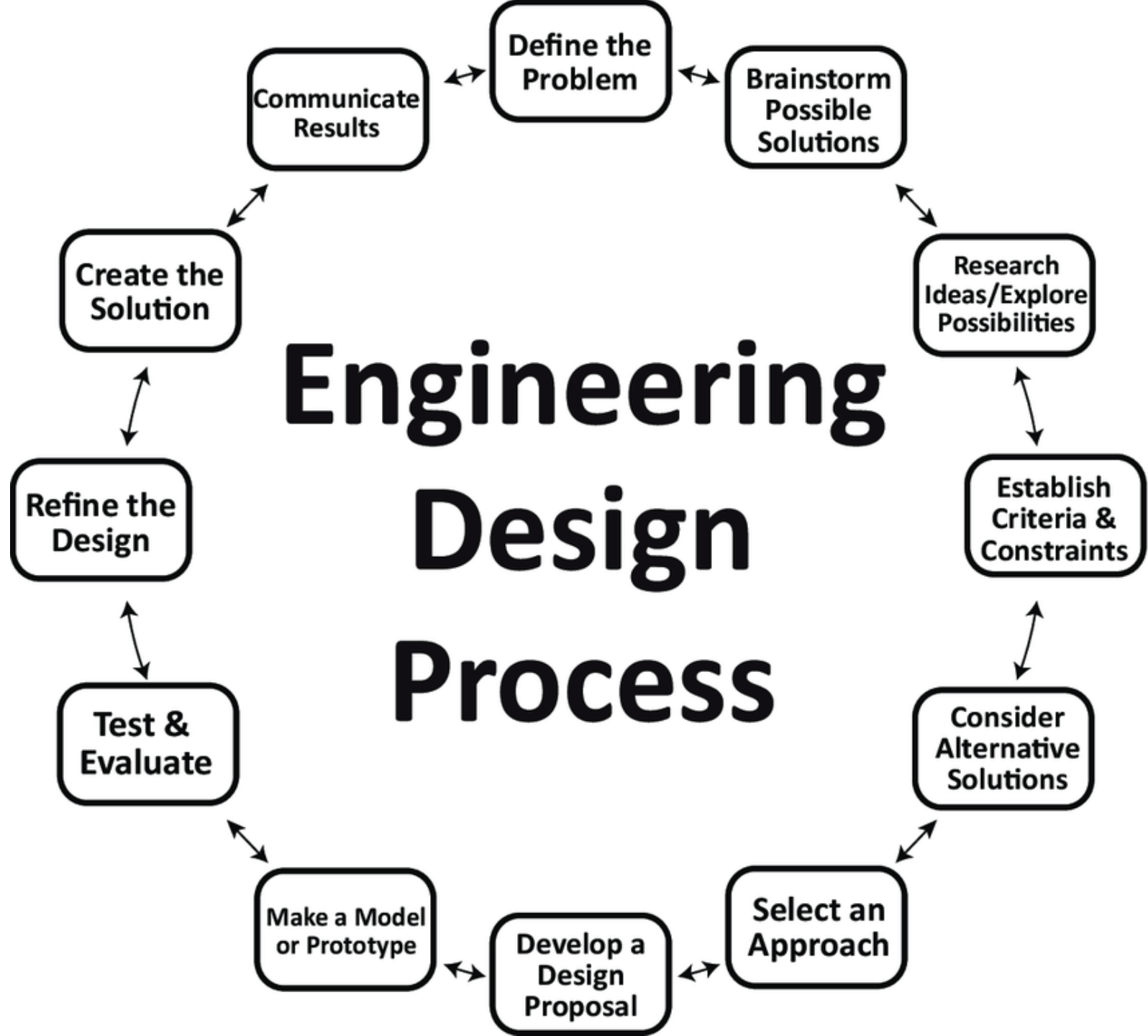
Design thinking brings together what is desirable from a human point of view with what is technologically feasible and economically viable.



Desirability: What makes sense to people and for people?

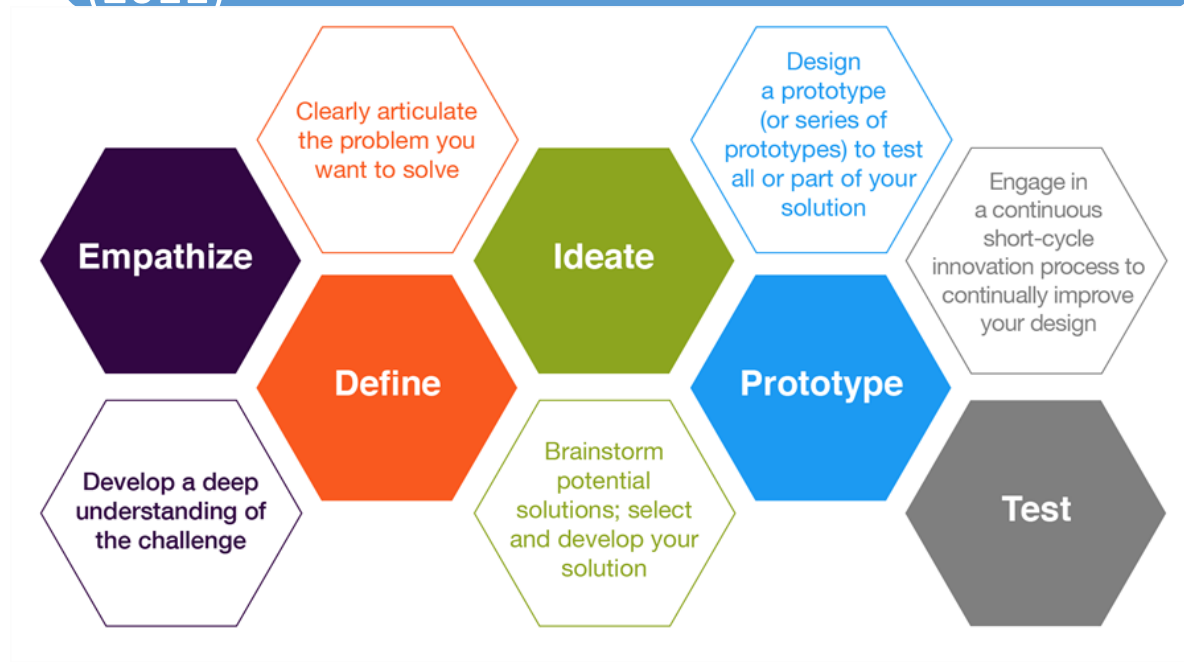
Feasibility: What is technically possible within the foreseeable future?

Viability: What is likely to become part of a sustainable business model?

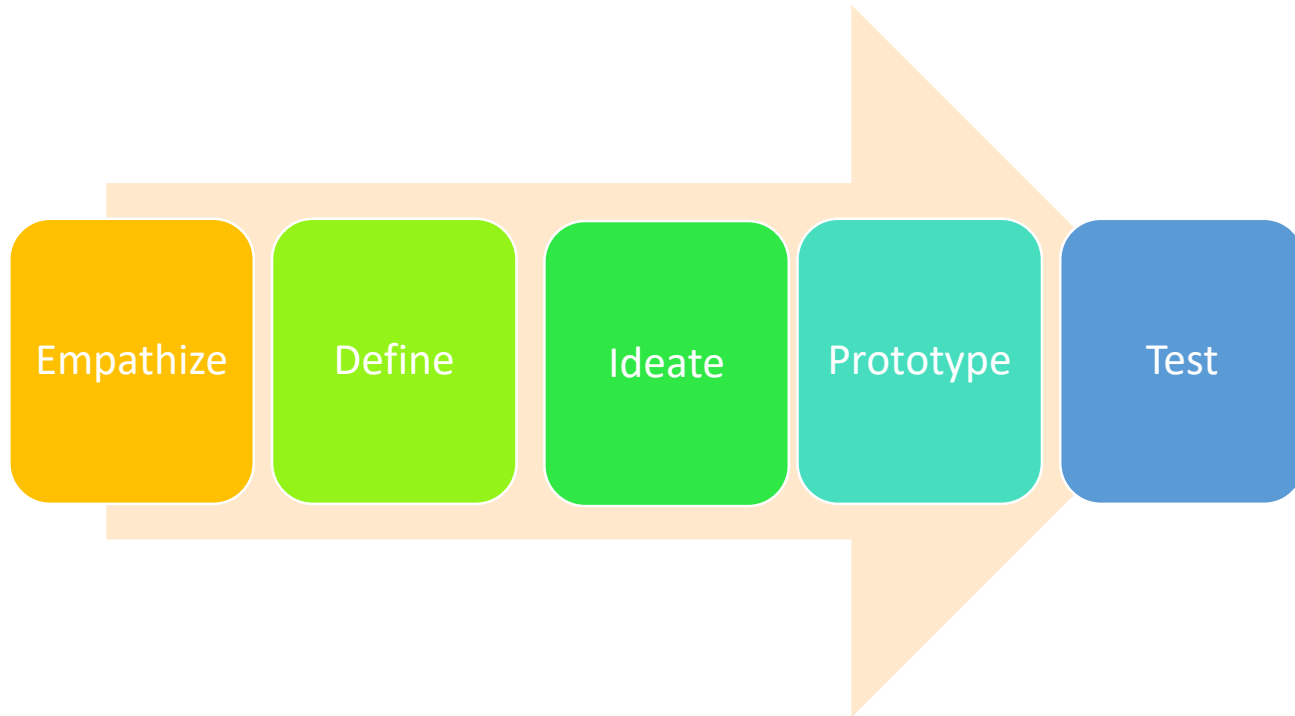


Design Thinking

Design Thinking is a more creative and user centered approach to problem solving than traditional design methods-Baeck & Gremett (2011)



Phases of Design Thinking



Empathize

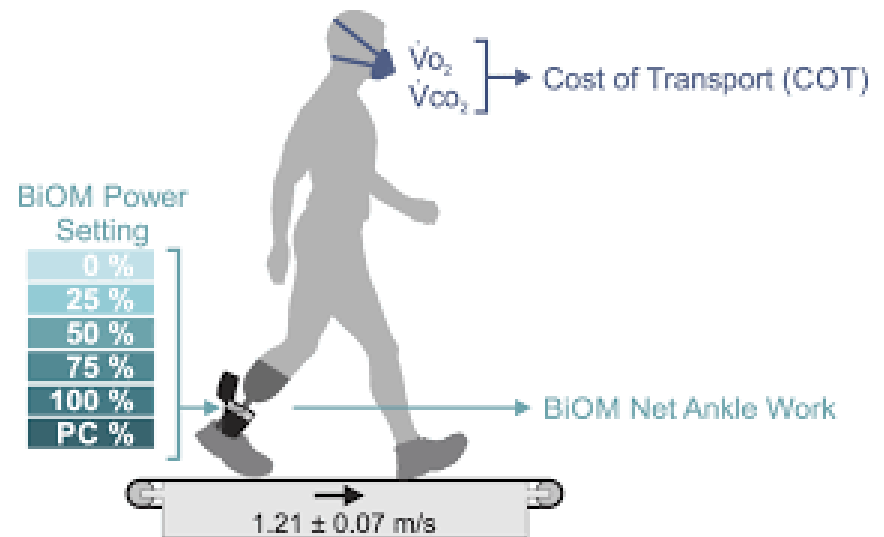


Learn about the people for whom you are designing.

Conduct research to develop a deeper understanding of your users.

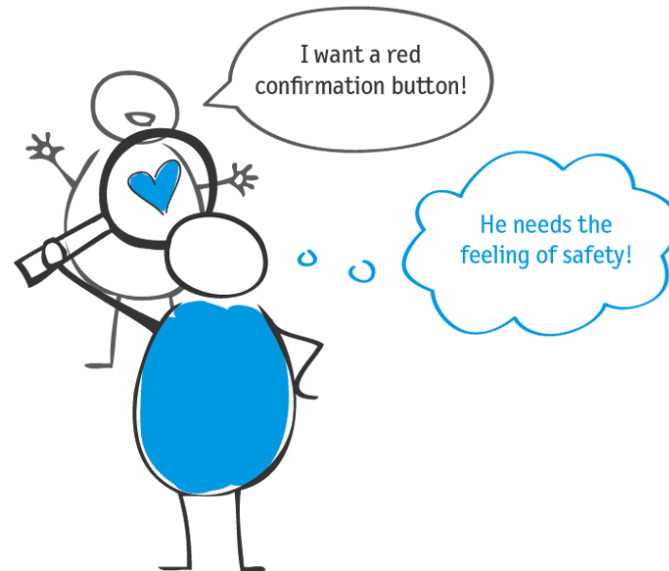


Example: Below-knee Amputees



Define

- Create a point of view that is based on user needs and insights



Example

- Normalize metabolic cost of amputees

Ideate



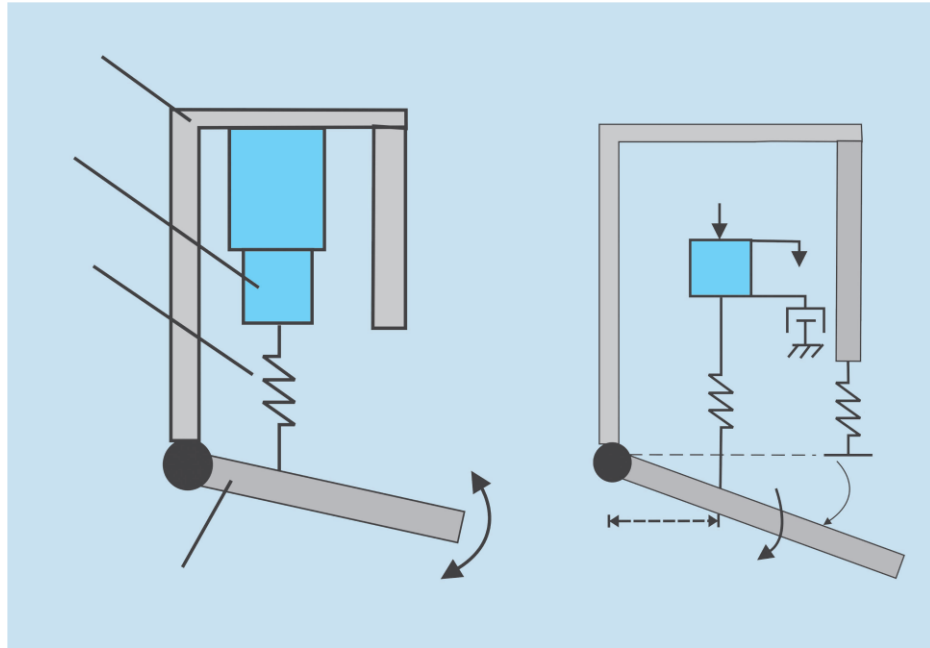
Brainstorm and come up with as many creative solutions as possible.



Generate a range of potential solutions by giving yourself and your team total freedom.

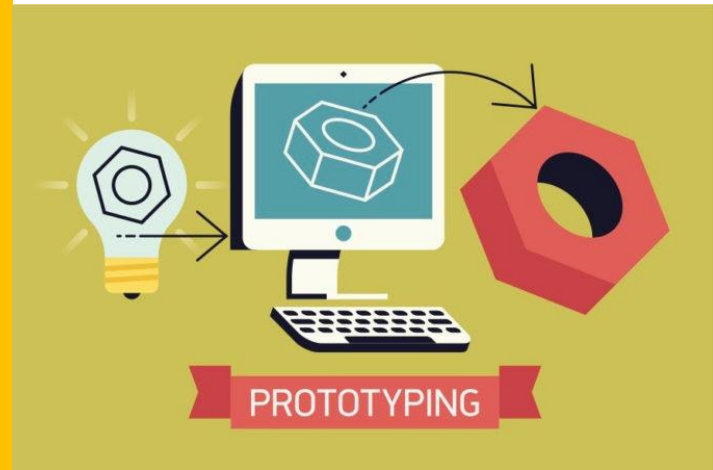
Example: Powered Prosthesis

- Hydraulic
- Pneumatic
- Electric
- Series Elastic

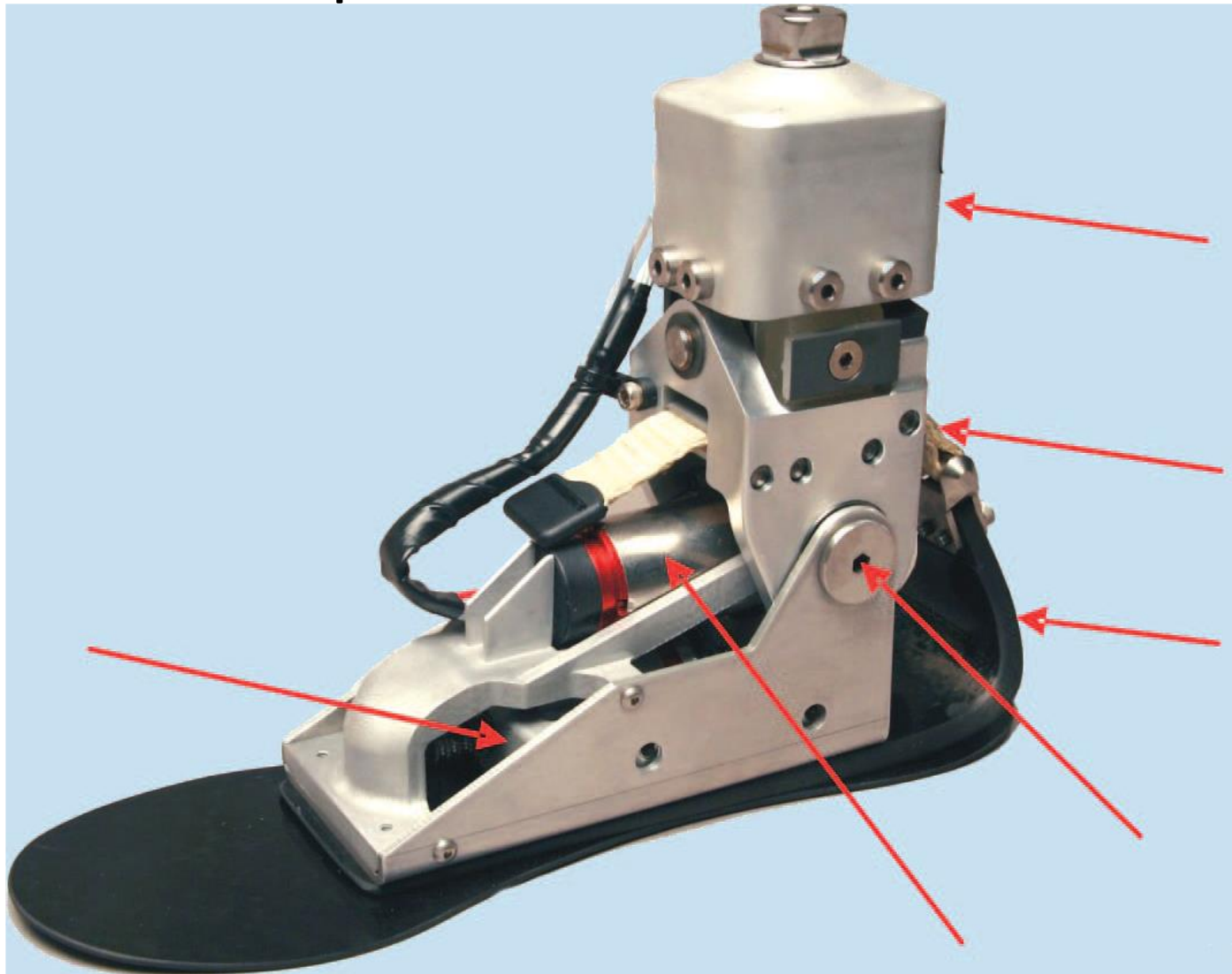


Prototype

- Build a prototype (or series of prototypes) to test your hypothesis.
- Creating a prototype lets designer see if they're on the right track, and it often sparks different ideas that you wouldn't have come up with otherwise.



Example: Powered Prosthesis



Test

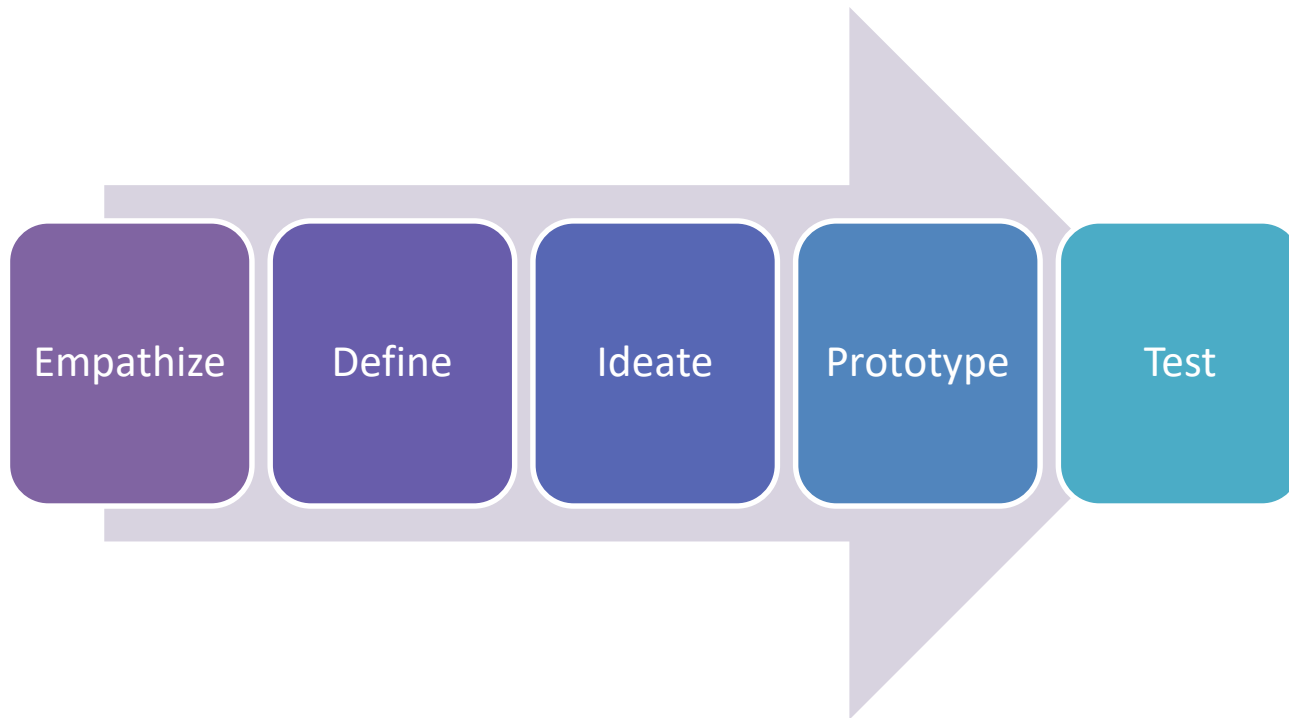
Return to your users for feedback.



Example



The Process-Design Thinking



human-centered

options-focused

possibility-driven

iterative